13.5 Asking Awkward Questions

WORKSHEET

The question

STEP II 2023, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Y is X_1 with probability p and X_2 with probability q. What is $\mathrm{E}(Y)$? What is $\mathrm{E}(Y^2)$?

STAGE 1 The variance of a mixture

Covers: Part (i)

Ask yourself Condition on which variable is chosen to find $\mathrm{E}(Y)$ and $\mathrm{E}(Y^2)$. Write $\mathrm{E}(X_i^2) = \sigma_i^2 + \mu_i^2$.

Pause and try Subtract $(\mathrm{E}(Y))^2$ and collect the μ terms, using $p - p^2 = pq$ and $q - q^2 = pq$.

STAGE 2 Independent randomisation

Covers: Part (ii)

Ask yourself What is $\mathrm{P}(B = 1)$? Are the n answers independent? So what is the distribution of Z_1?

Ask yourself Write "within 10% of the mean" in standard units. What happens as $n \to \infty$?

STAGE 3 One coin for everybody

Covers: Part (iii)

Pause and try Now one coin decides for all n dice. Given heads, what is the distribution of Z_2? Given tails? Use Part (i) for the mean and variance.

Ask yourself Where is most of the probability for Z_2? Is there much near $\frac{n}{2}$? Estimate how many standard deviations each hump is from $\frac{n}{2}$.

What's really going on

Randomised response

In Warner's scheme, each respondent answers "Do you belong to group A?" with probability p, and "Do you *not* belong to group A?" with probability $1-p$, choosing privately. If a fraction π belong to A, the probability of a "yes" is

$$\lambda = p\pi + (1-p)(1-\pi),$$

so from the observed fraction of yeses, $\hat{\lambda}$, we can estimate $\hat{\pi} = \frac{\hat{\lambda}-(1-p)}{2p-1}$ (provided $p \neq \frac{1}{2}$). Part (ii) of the question is exactly this, with each person's own die deciding. The price of privacy is precision: dividing by $2p-1$ inflates the variance, so a randomised survey needs more respondents than a direct one. But each person randomises independently, so the law of large numbers works. Part (iii) shows what goes wrong if the randomisation is shared: the uncertainty about the single coin never averages out.

Differential privacy

Randomised response is the ancestor of *differential privacy*, a mathematical definition of privacy introduced by Cynthia Dwork and colleagues in 2006: a procedure is private if adding or removing any one person's data changes the probability of any output by at most a small factor. Technology companies use it to collect usage statistics, and the United States Census Bureau used it for the 2020 census, adding carefully calibrated random noise to published figures so that no individual's answers could be reconstructed.

The law of total variance

The identity in part (i) is a special case of the *law of total variance*: $\operatorname{Var}(Y) = \operatorname{E}(\operatorname{Var}(Y \mid C)) + \operatorname{Var}(\operatorname{E}(Y \mid C))$, where C is the hidden choice. It underlies the *analysis of variance* (ANOVA), introduced by Ronald Fisher in the 1920s for agricultural experiments at Rothamsted, which splits the variation in crop yields into the part explained by fertilisers or varieties and the part left unexplained.

Try this next

1. **The twins.** Try STEP II 2005, Question 12 (step.maths.org/questions/05-s2-q12). It is a randomised-response puzzle in disguise: when does an answer tell you nothing about the coin?

2. **Estimating π.** In a Warner survey with $p = \frac{3}{4}$, 400 people answer and 160 say yes. Estimate the proportion π in group A, and find the approximate standard error of your estimate.

3. **Total variance.** Prove the law of total variance for a discrete hidden variable C with several values.

4. **Shared coins.** In part (iii), what is the probability that $Z_2 > \frac{n}{2}$? Explain why, for large n, it is close to $\frac{1}{2}$ even though Z_2 is almost never close to $\frac{n}{2}$.

Further reading

- **Stanley Warner, "Randomized response: a survey technique for eliminating evasive answer bias"**, *Journal of the American Statistical Association* 60 (1965).
- **Cynthia Dwork & Aaron Roth, *The Algorithmic Foundations of Differential Privacy*** (free online). The modern theory; the first chapters are accessible.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on surveys and their pitfalls.

The full guided solution is in Section 13.5:
https://beyondstep.keithmcnulty.org/ch13/awkward-questions.html